

Automotive Radar

Topic: A 77-GHz Hybrid TDM-MIMO Phased-Array Radar With 186-m Detection Range and 3-cm Range Resolution

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Ομάδα 04

Presentation Structure

- 1) Εισαγωγή στα Automotive Radars
- 2) Block Diagram of the hybrid radar transeiver.
- 3) Zig Zag tuning
- 4) Optimal High Switching Voltage
- 5) Block diagram of the proposed zig-zag PLL-based FMCW generator
- 6) Διανομή του LO chirp στα TX , RX channels
- 7) Αποτελέσματα
- 8) Βιβλιογραφία

Automotive Radar

A 77-GHz Hybrid TDM-MIMO Phased-Array Radar With 186-m Detection Range and 3-cm Range Resolution

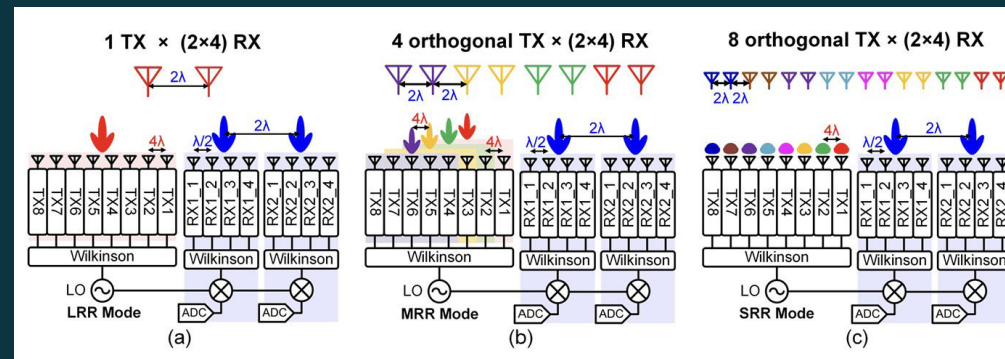
Architecture Phased array: Long Range Radar



Architecture TDM – MIMO : Short Range Radar



Hybrid radar → LRR / MRR / SRR → 186 m · 3 cm



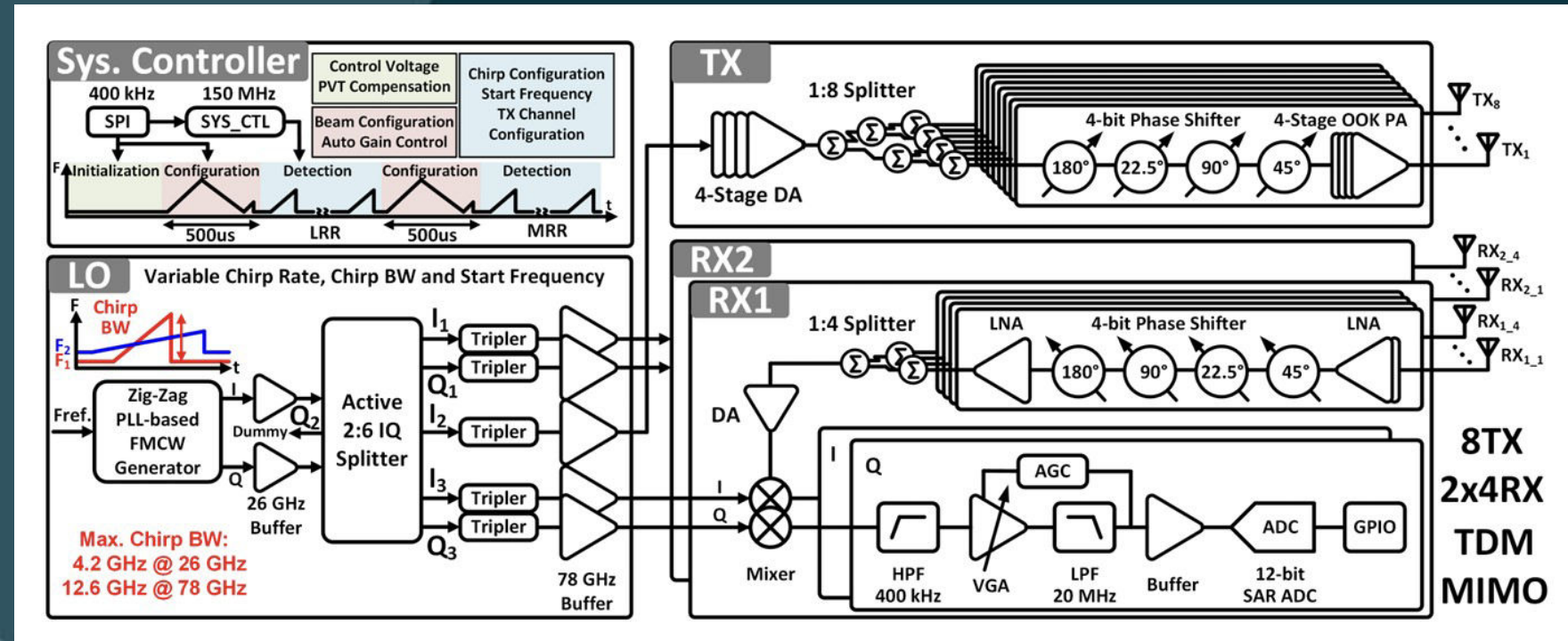
Block diagram of the proposed hybrid TDM-MIMO phased-array radar transceiver.

1) System Controller

2) LO Chirp Generation

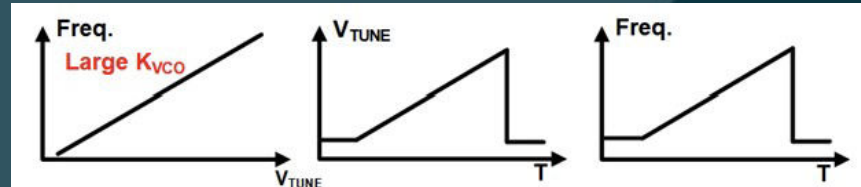
3) Tx Beamforming Path

4) Rx Baseband Path

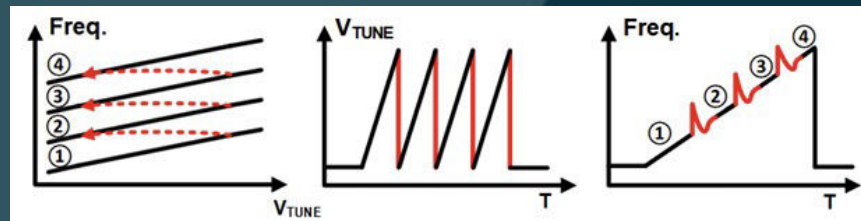


Πρόταση ZIG – ZAG tuning

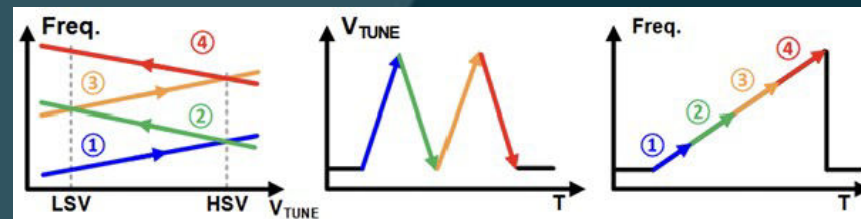
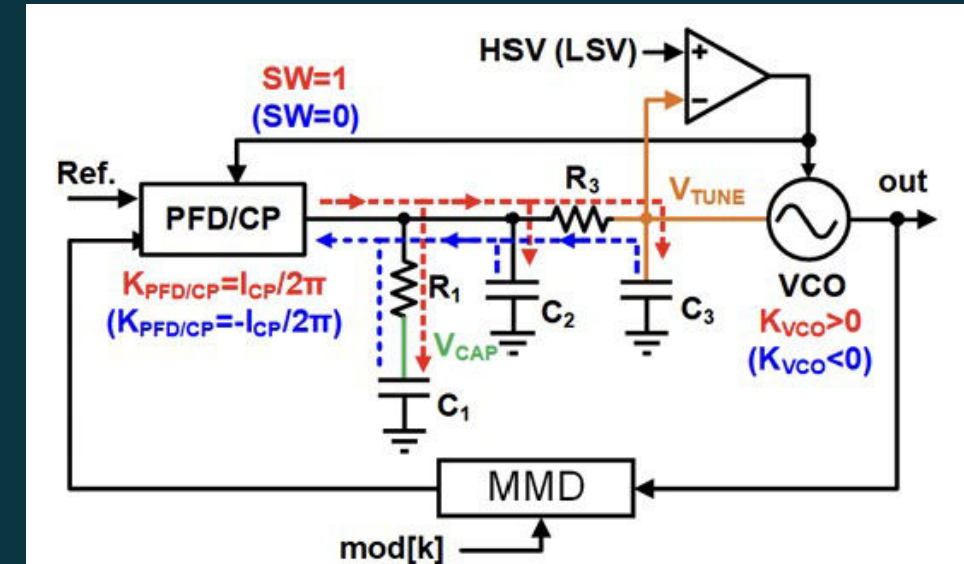
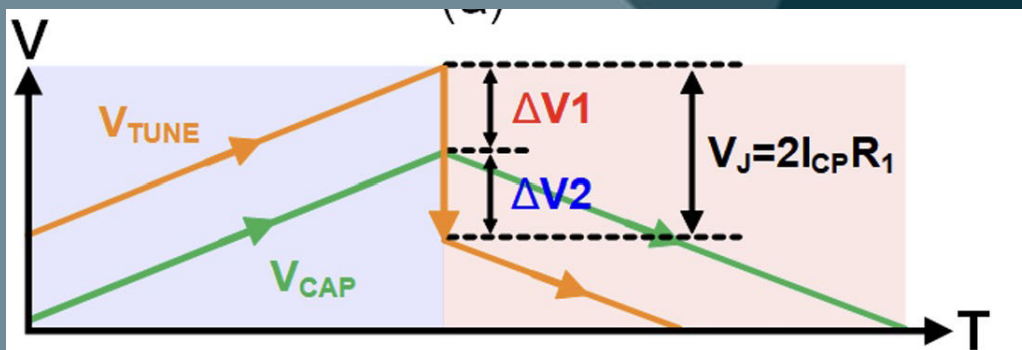
Single band chirp



Multi – band chirp

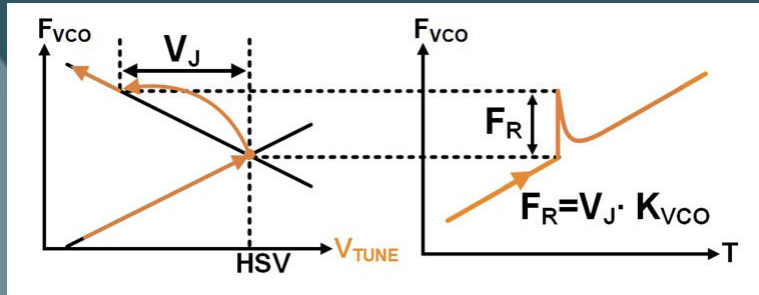


Zig – Zag subband chirp

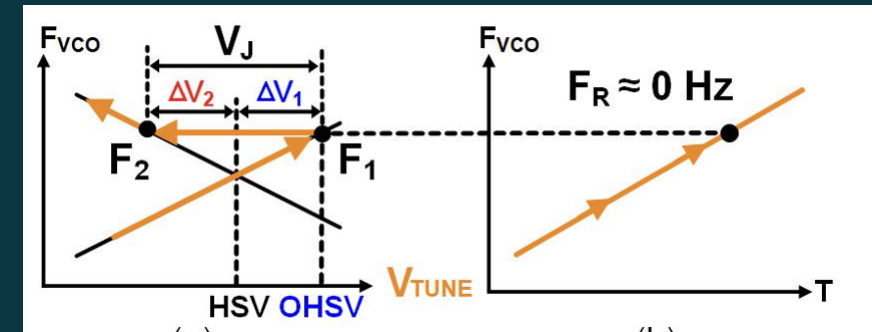
Direction of I_{CP}Voltage jump on V_{TUNE} 

Direction of I _{CP}	Voltage jump
→	$V_{CAP} + \Delta V_1$
←	$V_{CAP} - \Delta V_2$

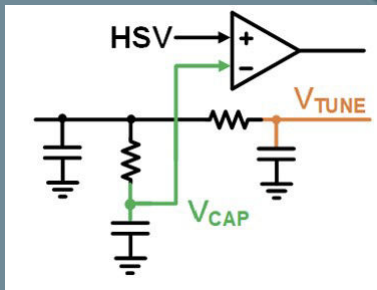
- V_J on V_{TUNE} causes frequency ripple :



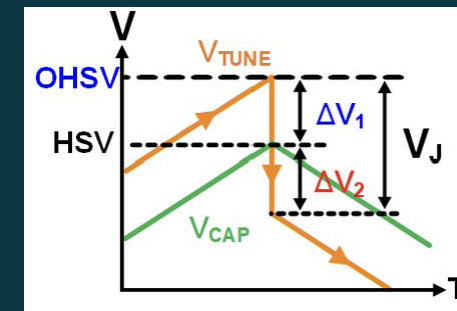
- F_R can be eliminated when switching voltage is OHSV :



- Detecting V_{CAP} to implement subband control :



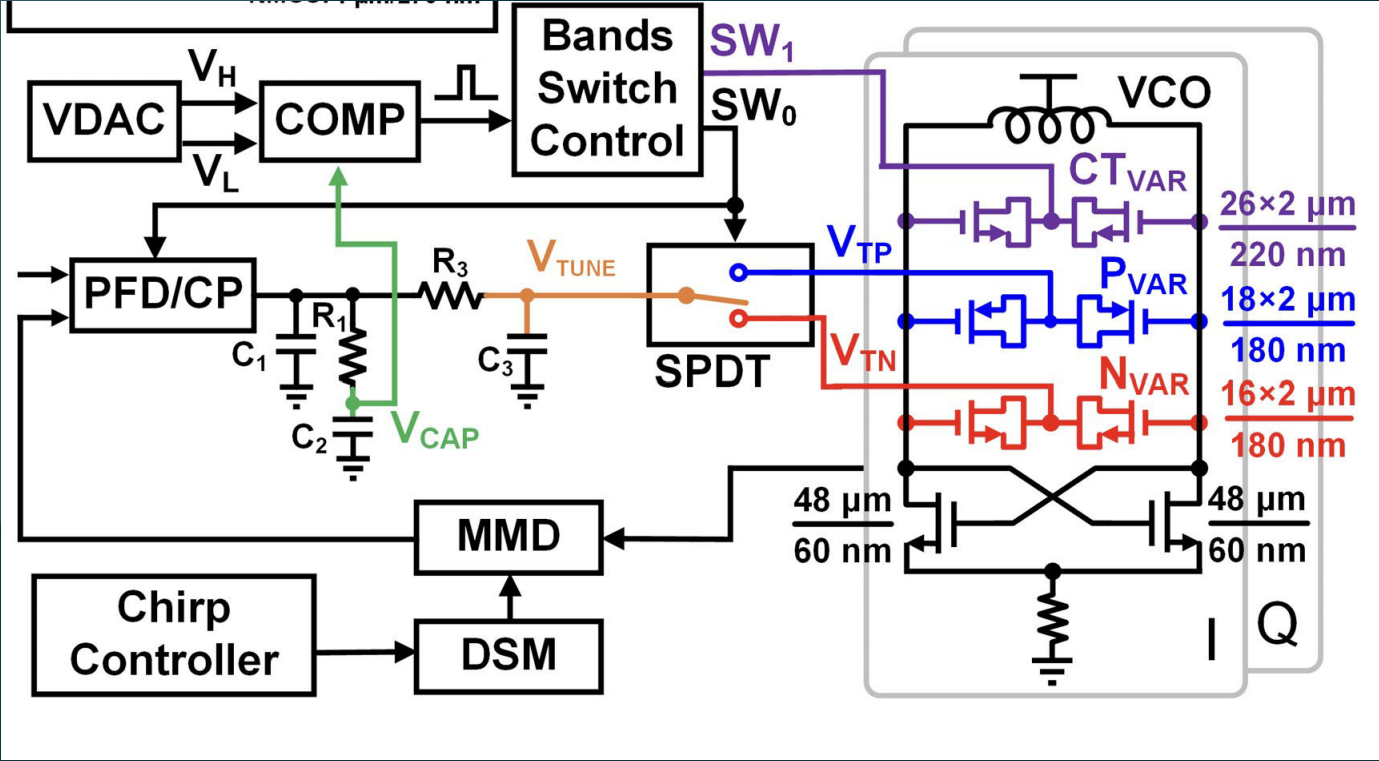
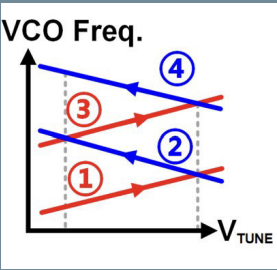
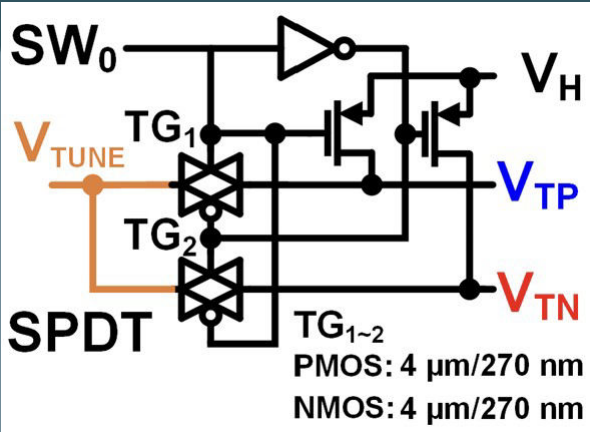
- Corresponding V_{CAP} and V_{TUNE} waveforms during switching :



$$\text{OHSV} = \text{HSV} + \Delta V_1 \approx \text{HSV} + I_{CP} \cdot R_1.$$

Block diagram of the proposed zig-zag PLL-based FMCW generator.

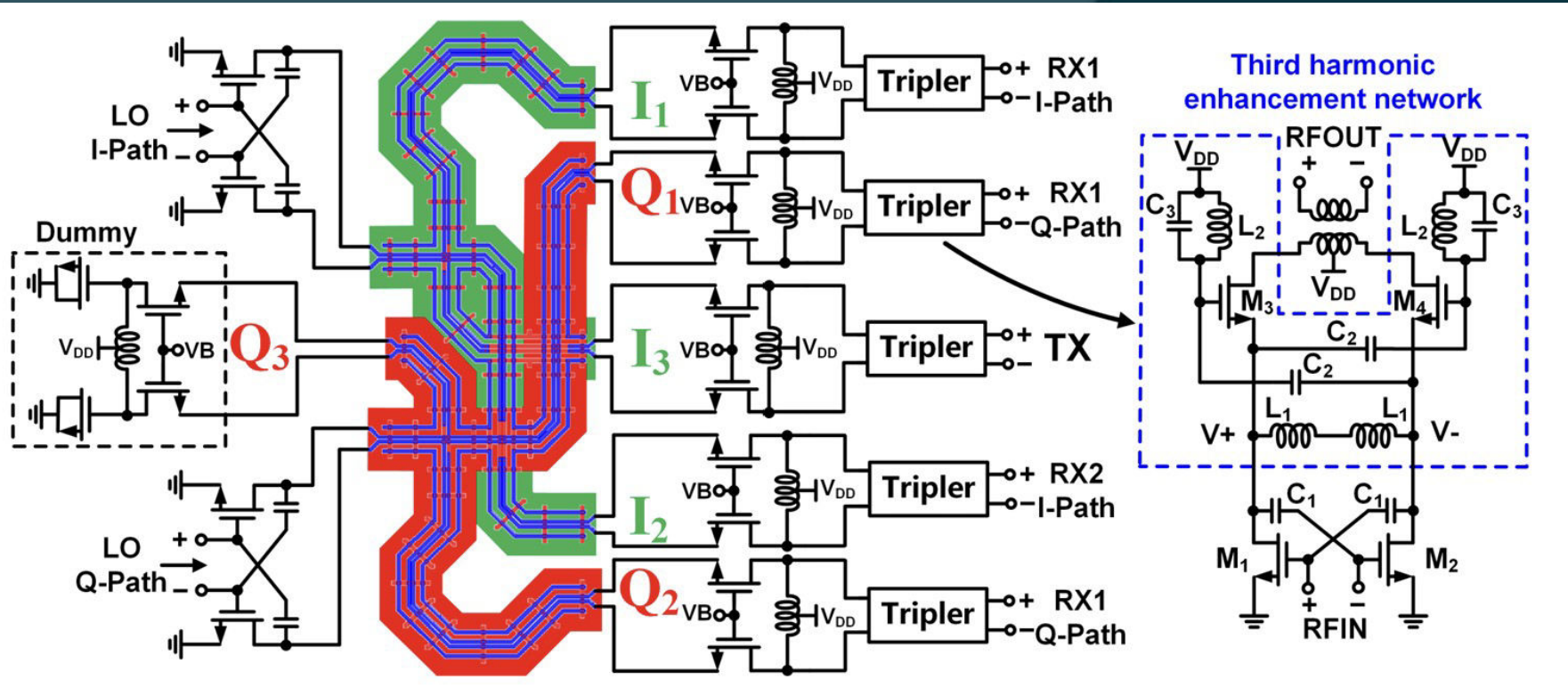
How VCO accomplece opposite polarity?



- 1) P – varactor , N – varactor
- 2) SPDT διακόπτης
- 3) SW₀ ελεγκτής
- 4) Phase – Frequency Detector / Charge Pump

Band	SW ₀	SW ₁	V _{TN}	V _{TP}
①	0	0	V _{TUNE}	V _H
②	1	0	V _H	V _{TUNE}
③	0	1	V _{TUNE}	V _H
④	1	1	V _H	V _{TUNE}

Διανομή του LO chirp στα TX , RX channels

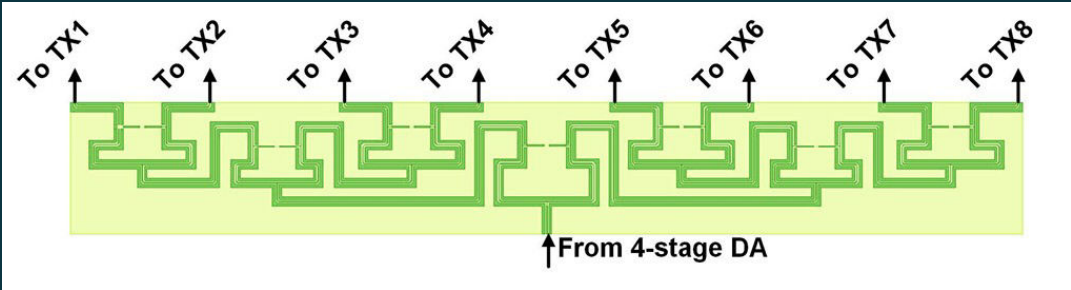


2:6 Active I/Q splitter

GCPW

Frequency Tripler

1:8 Wilkinson



Comparison of different LPFE topologies.

	This work	[5] ISSCC 2018	[33] ISSCC 2019	[34] JSSC 2021	[35] JSSC 2023	[16] JSSC 2020	[36] ISSCC 2019	[37] ISSCC 2021
Process	55-nm CMOS	45-nm CMOS	65-nm CMOS	40-nm CMOS	65-nm CMOS	65-nm CMOS	28-nm CMOS	45-nm CMOS
Integration	Front-ends, FMCW Gen., ABB, ADC	Front-ends, FMCW Gen., ABB, ADC, DBB	Front-ends, FMCW Gen., ABB, ADC, DBB	Front-ends, FMCW Gen., ABB, ADC	Front-ends, FMCW Gen., ABB, ADC	Front-ends, FMCW Gen., ABB, ADC	Front-ends, ABB, ADC, DBB	Front-ends, FMCW Gen., ABB, ADC, DBB
Structure	8TX 2x4RX	3TX 4RX	1TX 1RX	3TX 8RX	2TX 8RX	2TX 3RX	12TX 8x2RX	3TX 4RX
Modulation	FMCW	FMCW	FMCW	FMCW	FMCW	FMCW	PMCW	FMCW
Max. Range (m)	186	-	-	250	38.5	8.9	290	-
Range Resolution (m)	0.03	0.038*	0.038*	0.15*	0.021*	0.05	0.075	0.03*
Freq. (GHz)	77/79	77/79	77/79	77	77/79	77/79	77/79	76-81
Multi-band Chirp	Yes	No	No	No	No	No	No	No
Max. Chirp BW (GHz)	12.6	4	4	1	7.2	4	2	5
Max. Chirp Rate (MHz/μs)	545	100	340	40	312	133	-	266
Chirp Linearity (%@MHz/μs)	0.007@18.8 0.07@18.8** 0.30@545	0.06@100	0.02@58*** 0.06@112*** 0.20@340***	-	0.006@9.21 0.32@312	0.003@13.3 0.11@133	-	0.03@266
TX EIRP (dBm)	32 (Max.), 26 (Min.)	-	-	-	-	24.7 (Max.), 20.2 (Min.)	-	19 (Max.), 15 (Min.)
P_{OUT} (dBm)	5.73 (Max.), 4.83 (Min.)	13.6 (Max.), 10.8 (Min.)	13	14.1 (Max.), 11 (Min.)	15.5 (Max.), 12 (Min.)	13.5 (Max.), 9 (Min.)	19.6 (Max.)	13 (Max.), 12 (Min.)
RXFE CG (dB)	26.5 (Max.), 16 (Min.)	-	20	-	26 (Max.), 20 (Min.)	5	-	-
NF (dB)	16.3 (Min.), 27 (Max.) @2 MHz	13 (Min.), 18 (Max.) @NA	13@NA	8.7 (Min.), 14 (Max.) @10 MHz	11.2 (Min.), 26 (Max.) @10 MHz	15.3 (Min.), 18 (Max.) @600 kHz	16@NA	12 (Min), 13.8 (Max.) @NA
IF BW (MHz)	0.4-20	15	-	-	0.1-20	0.2-3	250-2000	20
PN@1MHz (dBc/Hz)	-91	-91	-91	-91	-90	-87.4	-110	-95
Equivalent Area (mm²)*****	1.05	1.83	2.3	1.925	2.1	1.22	0.37	-
Equivalent Power (mW)*****	105	292	700	133	223	154	78	292
Measured Temperature (°C)	65	-40 to 125	up to 150	-40 to 150	-45 to 125	-	up to 125	up to 140

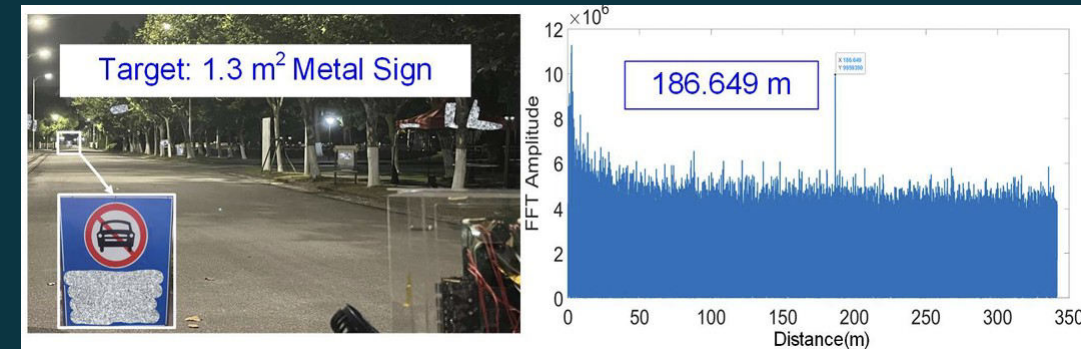
*Theoretically calculated value.

**Multi-band chirp measurement

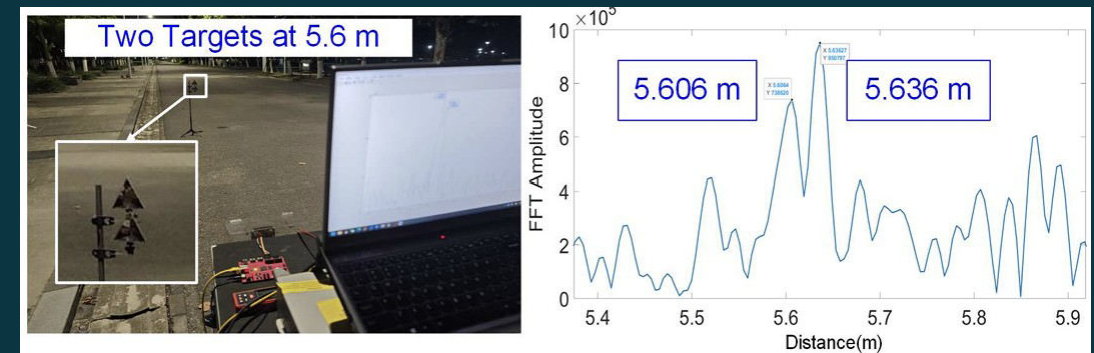
***Peak FM error.

*****The average area and consumption of each virtual channel (M TX + N RX means M $\times$ N virtual channels.)

Maximum detection range in LRR mode



Range resolution in SRR mode



Βιβλιογραφία

- 1) A 77 GHz Hybrid TDM – MIMO Phased – Array Radar With 186 - m Detection Range and 3 – cm Range Resolution. IEEE Journal of Solid – State Circuits, Vol 60 No 12
DECEMBER 2025
- 2) A High Linear PLL – based FMCW Frequency Synthesizer with 42 – kHz rms FM Error and 1.2 – GHz Chirp Bandwidth. IEEE Wireless and Microwave Technology Conference WAMICON.
2024.

Σας Ευχαριστώ.

Ερωτήσεις;